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Course Name : Compiler Design
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Short Answer test.

1) Grammar,

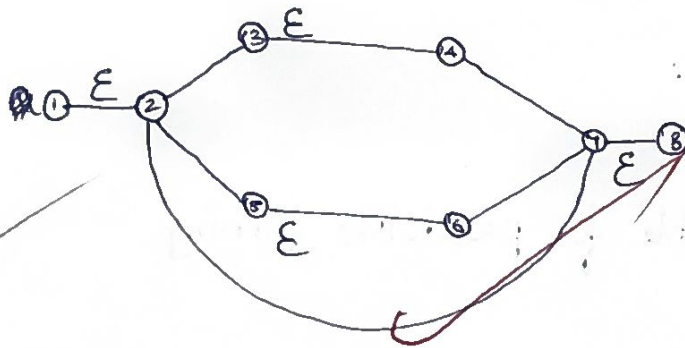
$$E \rightarrow E + E \mid id.$$

* Yes, it is an ambiguous because both the left side and the right side were in same terminal variable.

* So, it is an ambiguous.

2) Construct a Non-deterministic Finite Automaton.

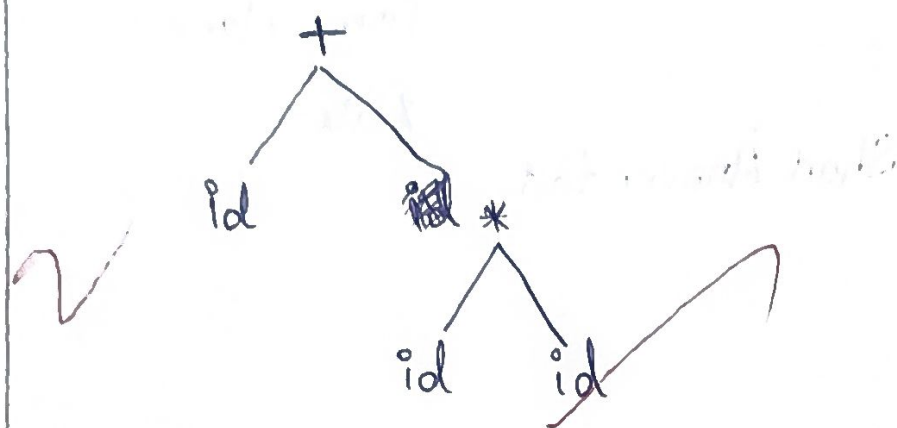
a) $a(ab)^*a$.



b) State transition diagram:

3) Construct a parse tree.

$id + id * id.$



4) Eliminate left recursion:

$A \rightarrow A\alpha \mid \beta.$

$A \rightarrow A\alpha \mid \beta$

$A \rightarrow A\alpha$

$A \rightarrow \beta A'$

$A \rightarrow A\alpha$

$A \rightarrow \alpha.$

5) Ambiguous and unsuitable for predictive parsing.

$S \rightarrow iEts \mid iETses \mid a.$

$C \rightarrow b.$

$S \rightarrow i(tses)$

Calculate first and follow:

First:

$$\text{First}(S) = \{a\}$$

$$\text{First}(B) = \{c\}$$

$$\text{First}(C) = \{b\}$$

$$\text{First}(C') = \{\epsilon\}$$

$$\text{First}(D) = \{E\}$$

$$\text{First}(E) = \{g\}$$

$$\text{First}(E') = \{E\}$$

$$\text{First}(F) = \{f\}$$

$$\text{First}(F') = \{E\}$$

Follow:

$$\text{Follow}(S) : \{\$, \}$$

$$\text{Follow}(B) : \{\$, g, f\}$$

$$\text{Follow}(C) : \{\$, g, f\}$$

$$\text{Follow}(C') : \{\$, g, f\}$$

$$\text{Follow}(D) : \{\$, g, f\}$$

$$\text{Follow}(E) : \{\$, g, f\}$$

$$\text{Follow}(E') : \{\$, g, f\}$$

$$\text{Follow}(F) : \{\$, g, f\}$$

$$\text{Follow}(F') : \{\$, g, f\}$$

1) Immediate left recursion:

Grammar:

$$S \rightarrow Sa | Sb | cd.$$

$$S \rightarrow Sa | Sb | cd.$$

$$S \rightarrow Sa$$

$$S \rightarrow Sb$$

$$S \rightarrow cd.$$

$$S \rightarrow bs'$$

$$S \rightarrow as'$$

$$S \rightarrow as' | bs' | \epsilon$$

$\therefore$ This Grammar contains Immediate left recursion.

8) For the grammar:

$E \rightarrow TE' , E' \rightarrow +TE' \mid \epsilon$, find $\text{First}(E')$.

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$\text{first } E' : \{+, \epsilon\}$.

9) For the grammar;

$$S \rightarrow (s)s \mid \epsilon$$

?

10)

(i) shift :

(ii) Reduce :

(iii) Rule :

(iv) accept :

Shift-Reduce Parsing:

11)

Grammar: $S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC \mid d$

Item: $[S' \rightarrow \cdot S]$

$\text{Goto}(I_0, S') \equiv I_1$

$S' \rightarrow S \cdot \checkmark$

$S \rightarrow \cdot CC$

$\text{Goto}(I_0, S) \equiv I_2$

$S \rightarrow C \cdot C$

$C \rightarrow \cdot cC \mid d$

$\text{Goto}(I_0, C) \equiv I_3$

$C \rightarrow c \cdot C \mid d$

$\text{Goto}(I_0, S) \equiv I_4$

$S \rightarrow CC \cdot \checkmark$

$C \rightarrow cC \cdot \checkmark$

12) Construct a first set for the non-terminal in

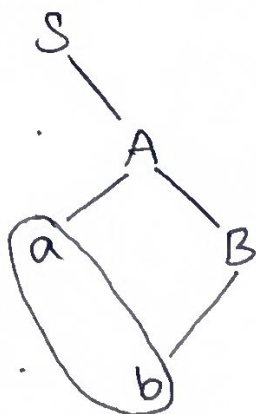
$S \rightarrow AB$

to accept: aabb.

$A \rightarrow a$

$B \rightarrow b$

Grammar: $S \rightarrow asb$



$\therefore aabb$ is accepted.

13) First & Follow:

first:

first (S) : {a}

first (A) : {a}

first (A') : {a}

first (B) : {b}

first (c) : {g}

follow:

follow (S) : {\$}

follow (A) : {\$, b}

follow (A') : {\$, b}

follow (B) : {\$, b}

follow (c) : {\$, b, g}.

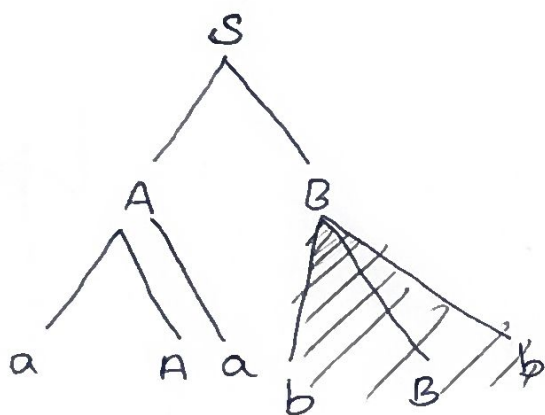
15) Grammar:

$S \rightarrow AB$

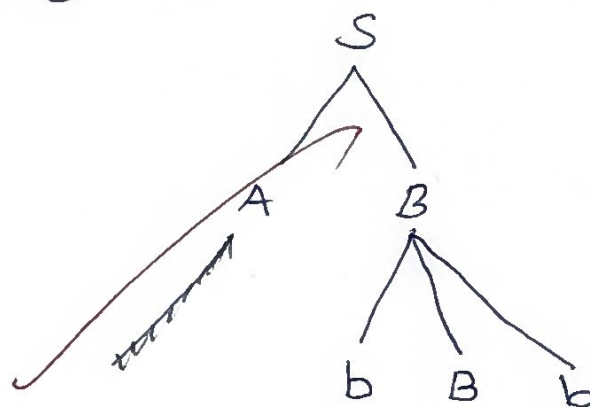
$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

LMD:



RMD



16)

$S \rightarrow AB$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

$S \rightarrow AB$

$A \rightarrow aA$

$A' \rightarrow a$

$B \rightarrow bB$

$B' \rightarrow b$

Follow:

follow (S) : {\$}

follow (A) : {\$, a}

follow (A') : {\$, a}

follow (B) : {\$, a, b}

follow (B') : {\$, a, b}.

Parsing technique :

Grammar :

String id + id * id

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' / \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' / \epsilon$$

$$F \rightarrow (E) / id.$$

$$First(E) : \{ (, id \}$$

$$First(E') : \{ +, \epsilon \}$$

$$First(T) : \{ (, id \}$$

$$First(T') : \{ *, \epsilon \}$$

$$First(F) : \{ (, id \}$$

$$Follow(E) : \{ \epsilon \}$$

$$Follow(E') : \{ \epsilon,) \}$$

$$Follow(T) : \{ +, \epsilon,) \}$$

$$Follow(T') : \{ +, id,) \}$$

$$Follow(F) : \{ *, +, \epsilon,) \}$$

(i)

Grammar	String	Action.
. \$	id + id * id \$	-
\$ E' T	id + id * id \$	$E \rightarrow TE'$
\$ T' F	id + id * id \$	$T \rightarrow FT'$
\$ id	+ id * id \$	$E \rightarrow (E)$
\$ E' T x	* id * id \$	$E' \rightarrow +TE'$
\$ T' F	id * id \$	$T \rightarrow FT'$
\$ id	* id \$	$F \rightarrow (E)$
\$ E' T	* id \$	$E \rightarrow TE'$
\$ T' F	* id \$	$T \rightarrow FT'$
\$ T' F *	id \$	$T' \rightarrow *FT'$
\$ id	id \$	$F \rightarrow (E)$
\$	\$	-

shift	(	id	+	*	)	\$
E	$E \rightarrow TE'$	$E \rightarrow TE'$				
E'			$E' \rightarrow +TE'$		$E' \rightarrow +TE'$	$E' \rightarrow +TE'$
T	$T \rightarrow FT'$	$T \rightarrow FT'$				
T'			$T' \rightarrow *FT'$	$T' \rightarrow *FT'$	$T' \rightarrow *FT'$	$T' \rightarrow *FT'$
F	$F \rightarrow (E)$	$F \rightarrow (E)$				

Q20

~~#include <stdio.h>~~

#include <stdio.h>

int main() {

int a=10, b=20;

int sum = a+b;

printf("%d", sum);

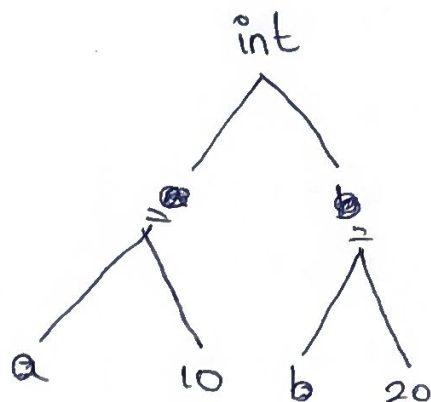
return 0;

}

a)

(i) $\langle \text{int} \rangle \langle a \rangle \langle = \rangle \langle 10 \rangle \langle , \rangle \langle b \rangle \langle = \rangle \langle 20 \rangle \langle ; \rangle$

(ii)

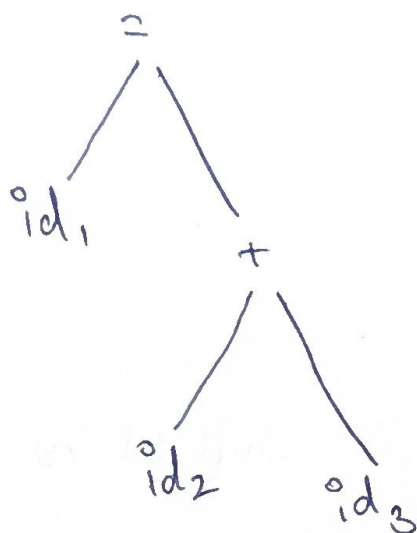


Logical Analyzer.



b) $\text{Sum} = a + b$

$\langle id_1 \rangle = \langle id_2 \rangle + \langle id_3 \rangle$



$E_1 = id_2 + id_3$

$E_2 = id_1$

c) $\text{Sum} = a + b$

$S \rightarrow id_1$

$E \rightarrow E + T \mid T$

$E \rightarrow +TE$

$E \rightarrow SE$

$T \rightarrow id$

8

4) Shift reduce.

$$E \rightarrow E + E$$

$$id_1 + id_2 * id_3.$$

$$E \rightarrow E * E$$

$$E \rightarrow (E)$$

$$E \rightarrow id$$

E	$id_1 + id_2 * id_3$	shift reduce
E	$E + id_2 * id_3$	Reduce by $E \rightarrow id$.
$E \rightarrow E$	$+ id_2 * id_3$	shift
$E \rightarrow E +$	$E id * id_3$	Reduce by $E \rightarrow id$
$E \rightarrow E + E$	$* id_3$	shift.
$E \rightarrow E *$	id_3	Reduce by $E \rightarrow id$.
$E \rightarrow E * E$	-	Reduce by $E \rightarrow E * E$.
$E \rightarrow (E)$	-	-